

Gowshick S

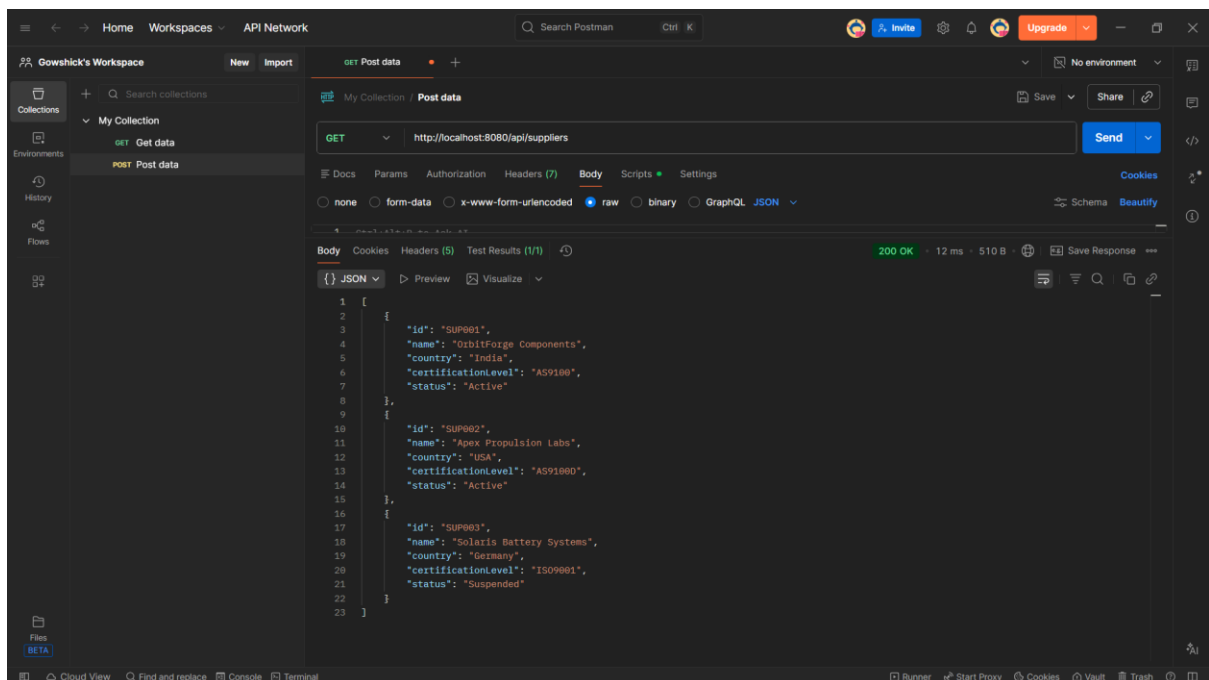
23BCE1200

AeroPassport:

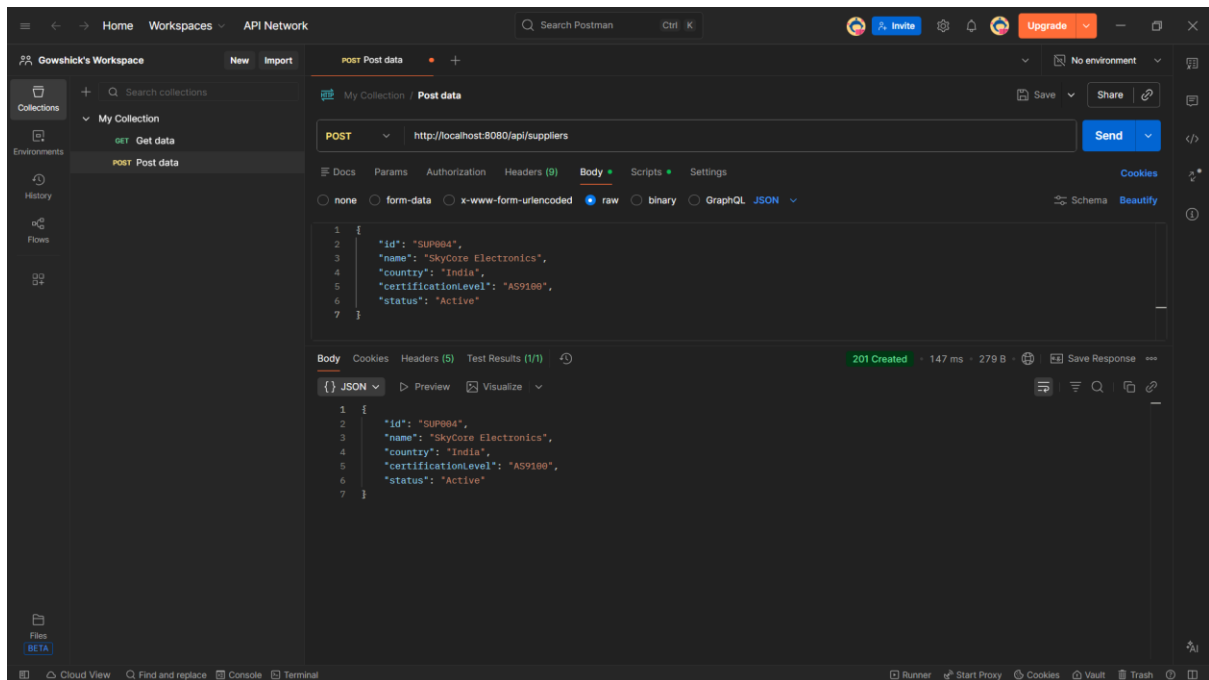
Backend initialization:



Data Seed:



Feature: Create Supplier

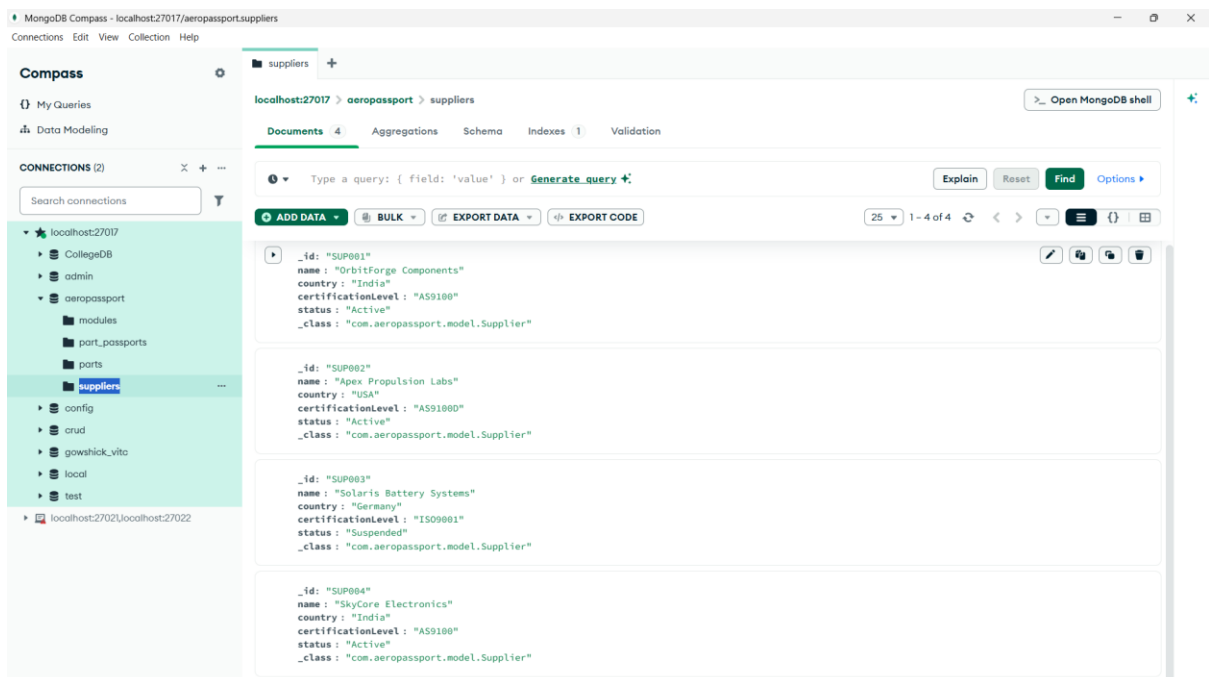


Supplier Created.

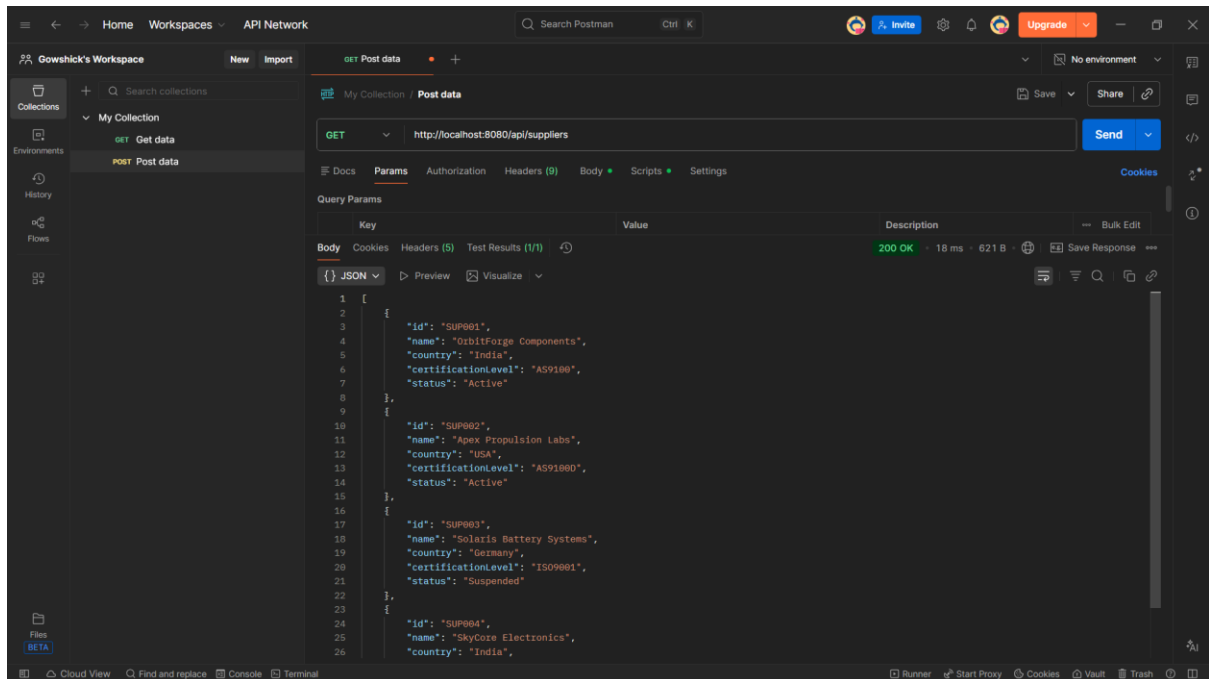
Spring Boot Verification:

```
2026-06-15T18:34:37.848+05:30 INFO 24296 --- [AeroPassport] [nio-8080-exec-5] c.a.controller.SupplierController
: [API] Creating supplier: Supplier(id=SUP004, name=SkyCore Electronics, country=India, certificationLevel=AS9100, statu
s=Active)
2026-06-15T18:34:37.859+05:30 DEBUG 24296 --- [AeroPassport] [nio-8080-exec-5] o.s.data.mongodb.core.MongoTemplate
: Saving Document containing fields: [_id, name, country, certificationLevel, status, _class]
```

MongoDB verification



Feature: Read Supplier



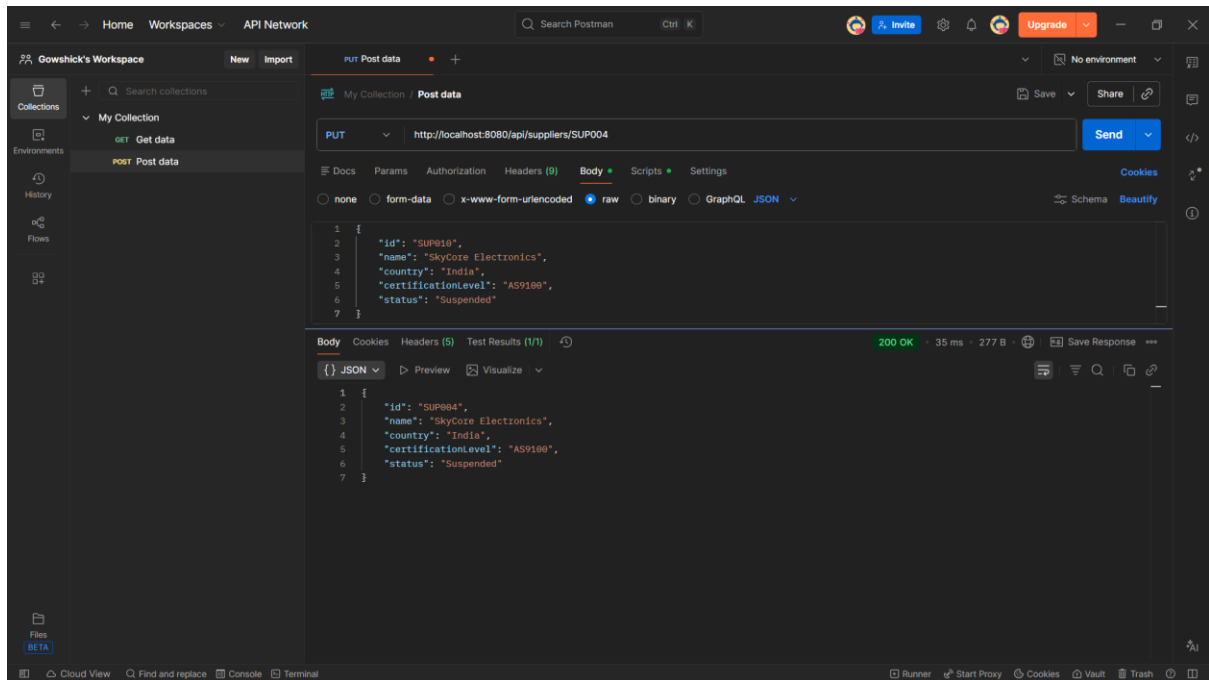
Suppliers' data read.

Spring Boot verification:

```
2026-06-15T18:39:42.209+05:30 DEBUG 24296 --- [AeroPassport] [nio-8080-exec-7] o.s.data.mongodb.core.MongoTemplate
: find using query: {} fields: Document{{{}} sort: null for class: class com.aeropassport.model.Supplier in collection: s
uppliers
```

Feature: Update Supplier info

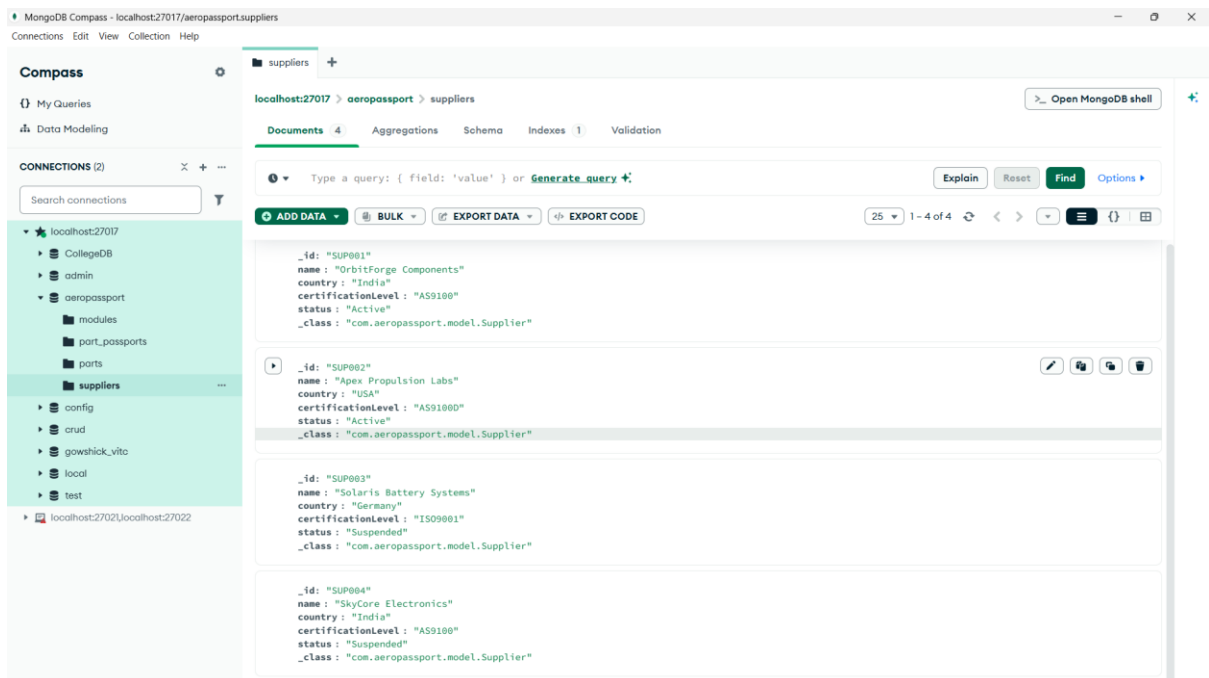
Update supplier status



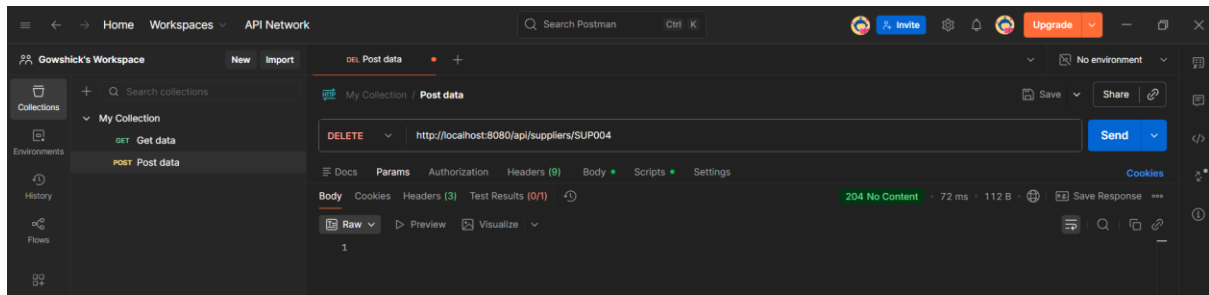
Supplier's status updated.

Spring Boot and DB Verification:

```
2026-06-15T18:42:00.830+05:30 DEBUG 24296 --- [AeroPassport] [io-8080-exec-10] o.s.data.mongodb.core.MongoTemplate
: findOne using query: { "id" : "SUP004" } fields: Document{{{}} for class: class com.aeropassport.model.Supplier in colle
ction: suppliers
2026-06-15T18:42:00.842+05:30 DEBUG 24296 --- [AeroPassport] [io-8080-exec-10] o.s.data.mongodb.core.MongoTemplate
: Saving Document containing fields: [_id, name, country, certificationLevel, status, _class]
2026-06-15T18:42:00.847+05:30 INFO 24296 --- [AeroPassport] [io-8080-exec-10] c.a.controller.SupplierController
: [API] Updated supplier: Supplier(id=SUP004, name=SkyCore Electronics, country=India, certificationLevel=AS9100, status
=Suspended)
```



Feature: Delete Supplier

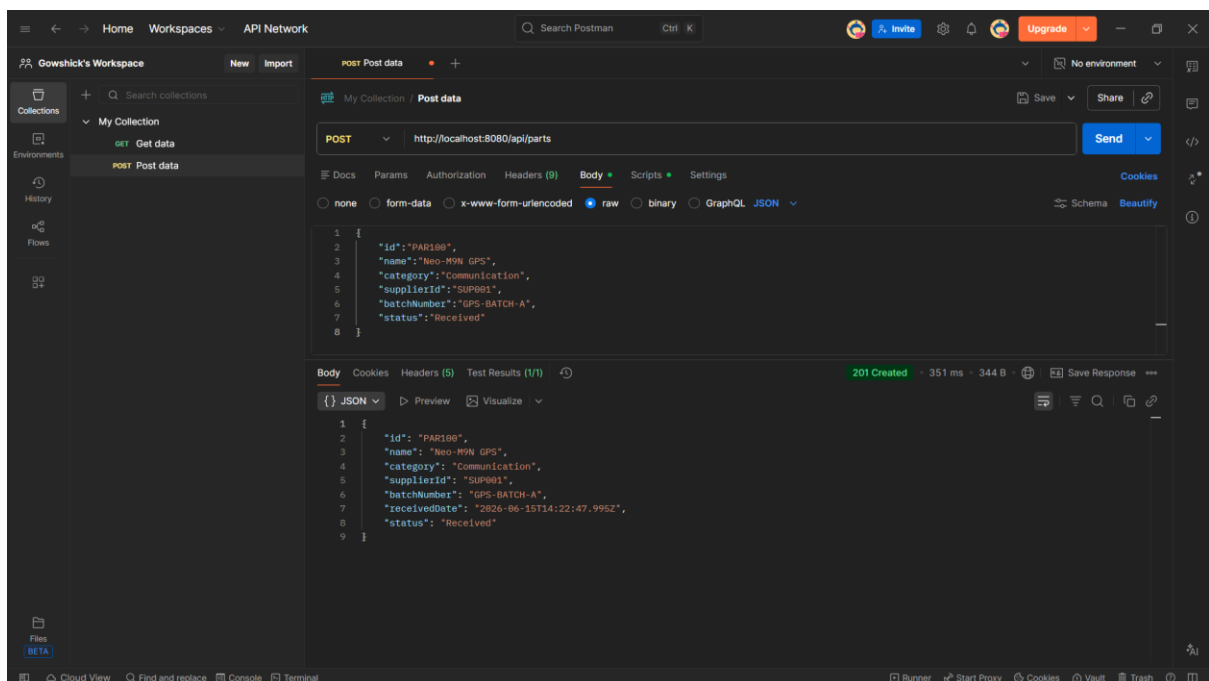


Supplier deleted.

Spring Boot Verification:

```
-SuspendedJ
2026-06-15T18:43:44.072+05:30 DEBUG 24296 --- [AeroPassport] [nio-8080-exec-4] o.s.data.mongodb.core.MongoTemplate
: Executing count: { "_id" : "SUP004" } in collection: suppliers
2026-06-15T18:43:44.109+05:30 DEBUG 24296 --- [AeroPassport] [nio-8080-exec-4] o.s.data.mongodb.core.MongoTemplate
: Remove using query: { "_id" : "SUP004" } in collection: suppliers.
2026-06-15T18:43:44.126+05:30 INFO 24296 --- [AeroPassport] [nio-8080-exec-4] c.a.controller.SupplierController
: [API] Deleted supplier with id: SUP004
```

Feature: Create and Verify “Part”



Part created.

Spring Boot Verification:

```
2026-06-15T19:52:47.981+05:30 INFO 24296 --- [AeroPassport] [nio-8080-exec-2] c.a.controller.PartController
: [API] Registering new part: Part(id=PAR100, name=Neo-M9N GPS, category=Communication, supplierId=SUP001, batchNumber=G
PS-BATCH-A, receivedDate=null, status=Received)
2026-06-15T19:52:48.040+05:30 DEBUG 24296 --- [AeroPassport] [nio-8080-exec-2] o.s.data.mongodb.core.MongoTemplate
: Saving Document containing fields: [_id, name, category, supplierId, batchNumber, receivedDate, status, _class]
2026-06-15T19:52:48.059+05:30 INFO 24296 --- [AeroPassport] [nio-8080-exec-2] com.aeropassport.service.PartService
: [AUDIT LOG] Registered flight hardware part: PAR100 - Neo-M9N GPS (Batch: GPS-BATCH-A)
[AUDIT LOG] Registered flight hardware part: PAR100 - Neo-M9N GPS (Batch: GPS-BATCH-A)
2026-06-15T19:52:48.087+05:30 DEBUG 24296 --- [AeroPassport] [nio-8080-exec-2] o.s.data.mongodb.core.MongoTemplate
: Saving Document containing fields: [_id, partId, certifications, flightReadiness, remarks, inspections, events, _class
]
2026-06-15T19:52:48.093+05:30 INFO 24296 --- [AeroPassport] [nio-8080-exec-2] com.aeropassport.service.PartService
: [AUDIT LOG] Created digital PartPassport PAS-PAR100 for part PAR100
[AUDIT LOG] Created digital PartPassport PAS-PAR100 for part PAR100
```

DB Verification of “Part” creation:

The screenshot shows the MongoDB Compass interface for the 'aeropassport' database, specifically the 'parts' collection. The left sidebar shows the database structure with 'parts' selected. The main panel displays three documents in the 'parts' collection:

- Document 1:** A failed battery pack.

```
{
  "_id": "PAR002",
  "name": "LC0-18650 Battery Pack",
  "category": "Power",
  "supplierId": "SUP003",
  "batchNumber": "B2026-BATT",
  "receivedDate": "2026-06-11T13:51:16.133+00:00",
  "status": "Failed",
  "_class": "com.aeropassport.model.Part"
}
```
- Document 2:** A received propulsor controller.

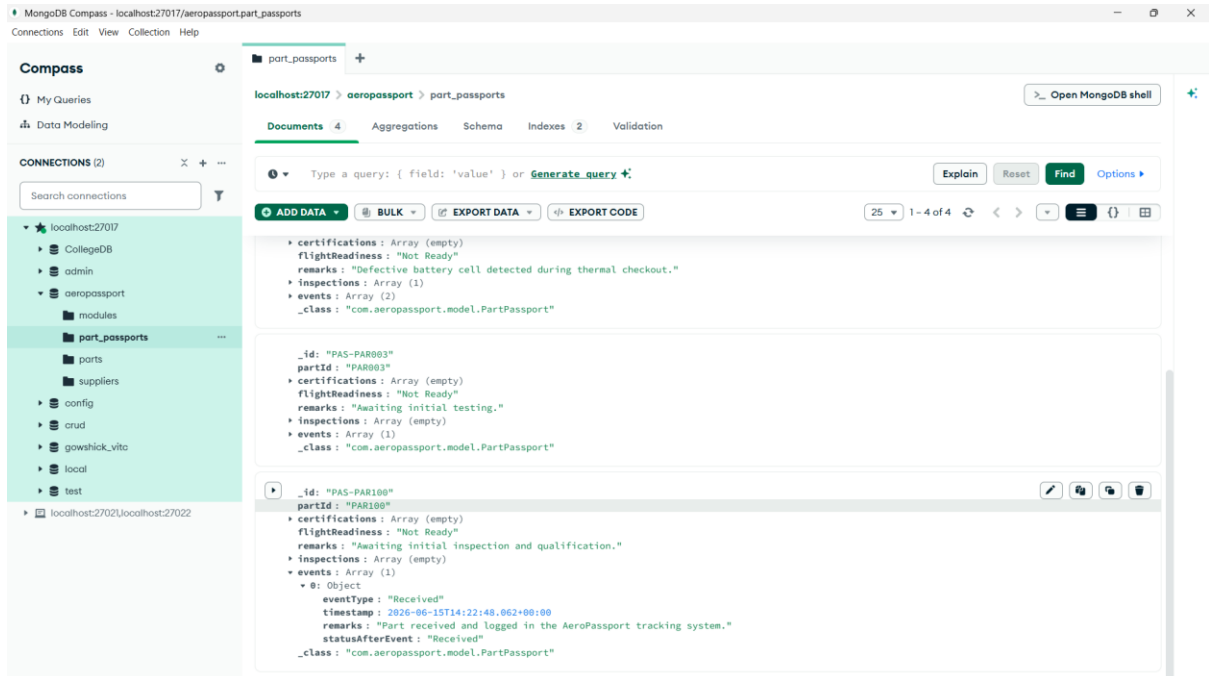
```
{
  "_id": "PAR003",
  "name": "Raptor Thruster Controller",
  "category": "Propulsion",
  "supplierId": "SUP002",
  "batchNumber": "B2026-PROP",
  "receivedDate": "2026-06-13T13:51:16.133+00:00",
  "status": "Received",
  "_class": "com.aeropassport.model.Part"
}
```
- Document 3:** A received Neo-M9N GPS.

```
{
  "_id": "PAR100",
  "name": "Neo-M9N GPS",
  "category": "Communication",
  "supplierId": "SUP001",
  "batchNumber": "GPS-BATCH-A",
  "receivedDate": "2026-06-15T14:22:47.995+00:00",
  "status": "Received",
  "_class": "com.aeropassport.model.Part"
}
```

Core Feature: Automatic Part Passport generation

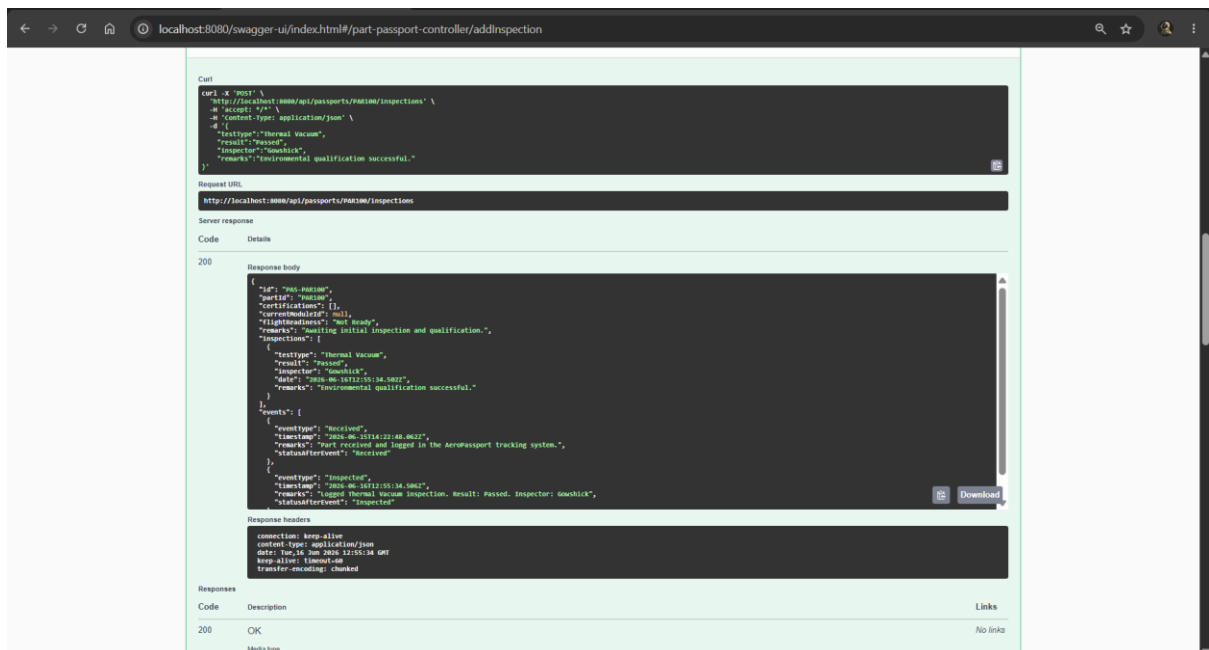
Verification

Immediately after Part creation:



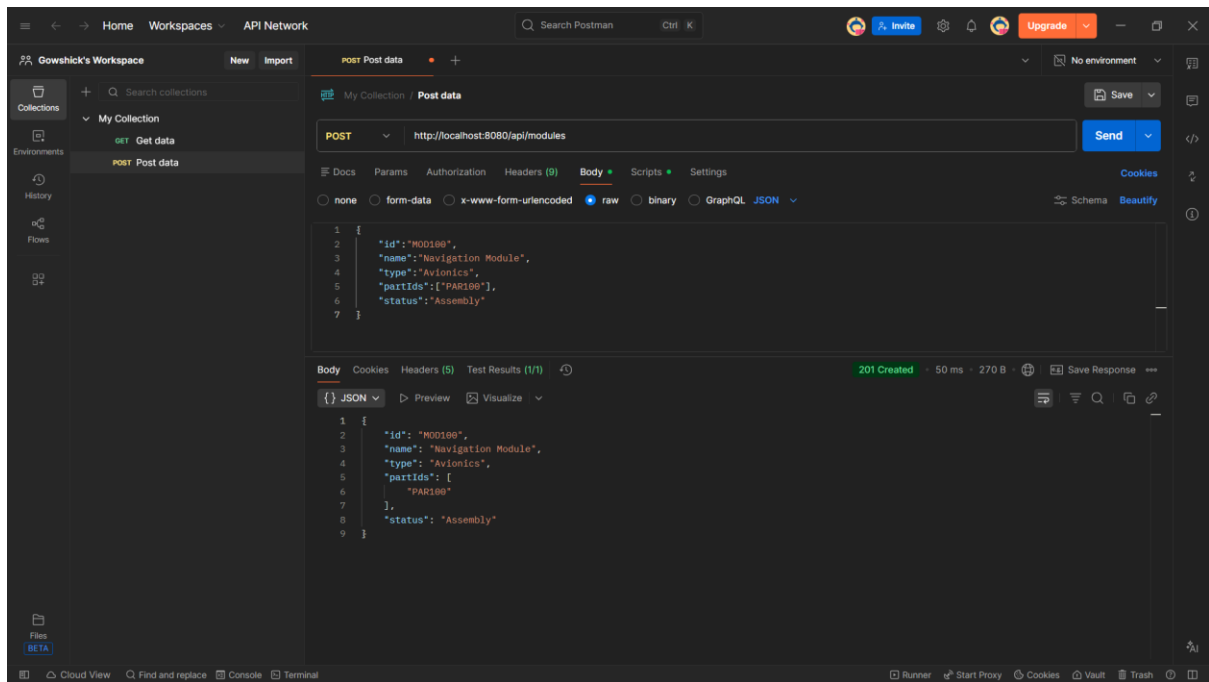
Passport is created and embedded Events Array contain: “Received” status

Swagger Verification of Embedded Inspection:



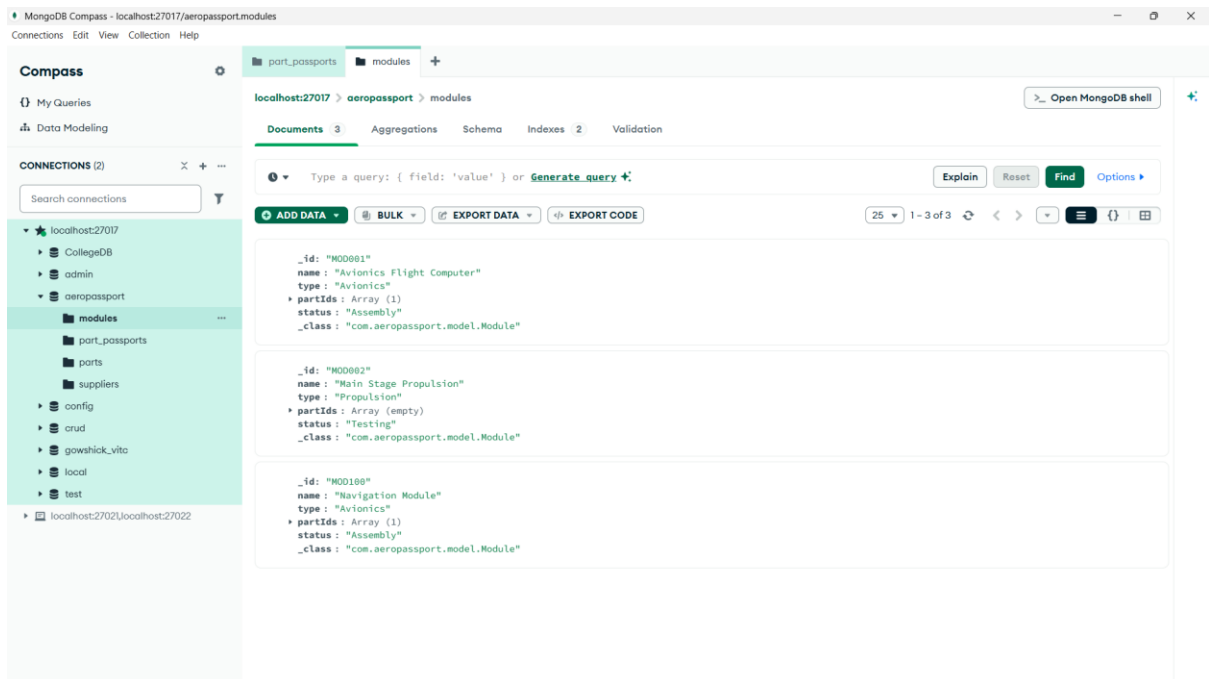
Inspections and Events created automatically.

Module Creation:



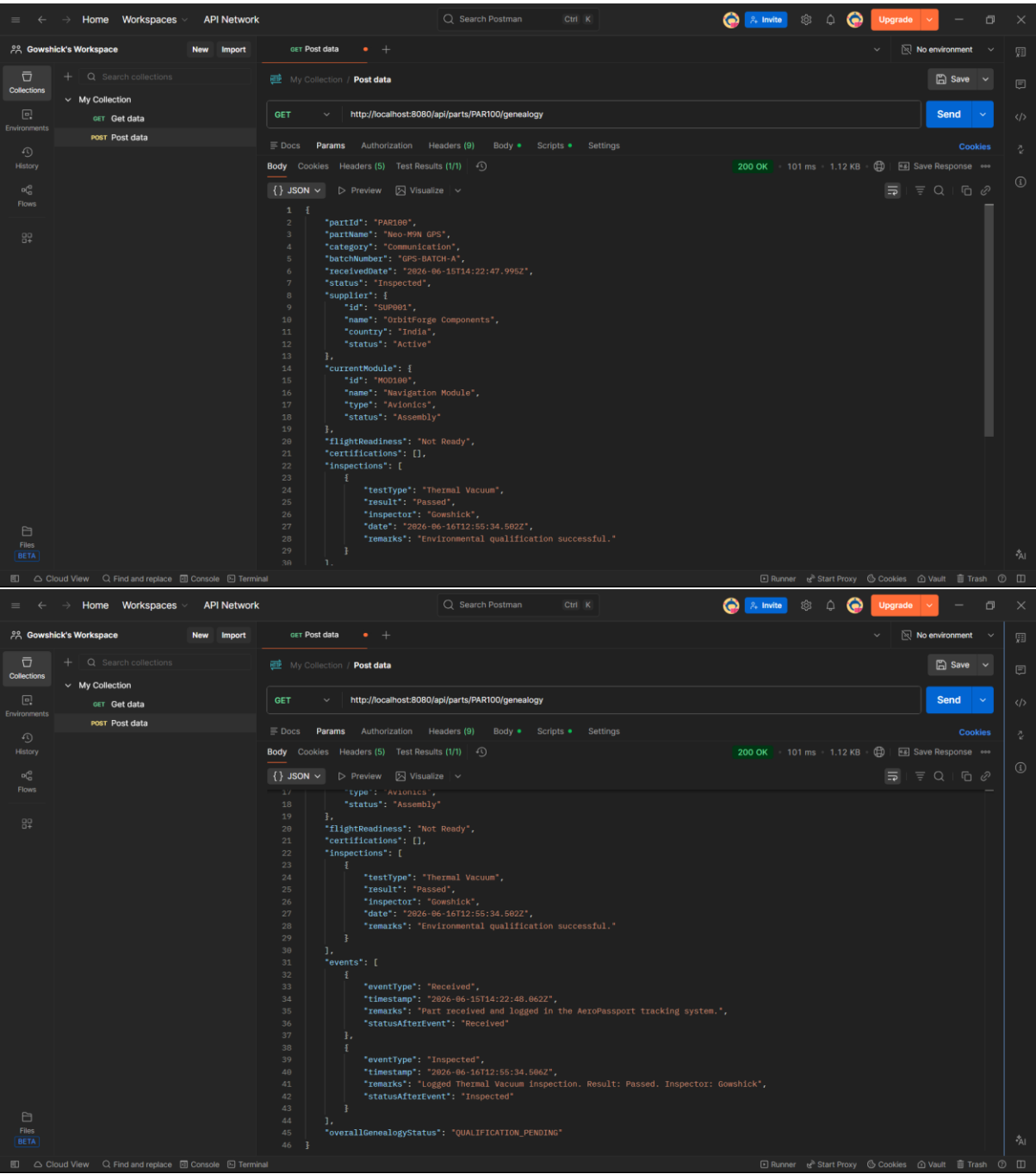
Module created.

DB and Spring Boot Verification:



```
2026-06-16T18:28:03.661+05:30 INFO 12076 --- [AeroPassport] [nio-8080-exec-2] c.a.controller.ModuleController
: [API] Creating subsystem module: Module(id=MOD100, name=Navigation Module, type=Avionics, partIds=[PAR100], status=Ass
embly)
2026-06-16T18:28:03.668+05:30 DEBUG 12076 --- [AeroPassport] [nio-8080-exec-2] o.s.data.mongodb.core.MongoTemplate
: Saving Document containing fields: [_id, name, type, partIds, status, _class]
```

Core Feature: Verify Hardware Genealogy Report



Report Generated.

The report contains all the data of:

- Part
- Supplier
- Module

Inspections (Embedded array)

Events (Embedded array)

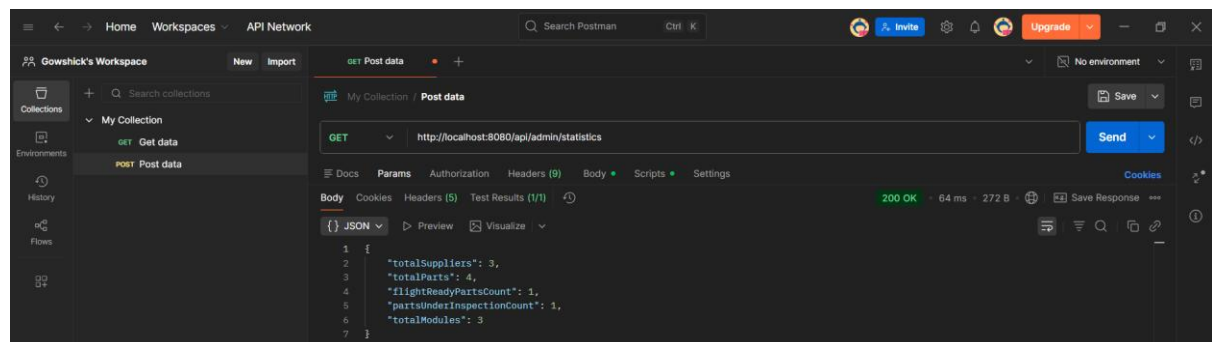
Readiness

Delivered Status

Spring Boot Verification:

```
2026-06-16T18:30:19.966+05:30 INFO 12076 --- [AeroPassport] [nio-8080-exec-5] c.a.controller.ReportController
: [API] Requested Hardware Genealogy Report for part ID: PAR100
2026-06-16T18:30:19.968+05:30 INFO 12076 --- [AeroPassport] [nio-8080-exec-5] com.aeropassport.service.ReportService
: [GENEALOGY REPORT] Generating traceability report for part ID: PAR100
2026-06-16T18:30:20.024+05:30 DEBUG 12076 --- [AeroPassport] [nio-8080-exec-5] o.s.data.mongodb.core.MongoTemplate
: Executing aggregation: [{ "$match" : { "_id" : "PAR100"}}, { "$lookup" : { "from" : "suppliers", "localField" : "supplierId", "foreignField" : "_id", "as" : "supplierList"}}, { "$lookup" : { "from" : "part_passports", "localField" : "_id", "foreignField" : "partId", "as" : "passportList"}}, { "$lookup" : { "from" : "modules", "localField" : "_id", "foreignField" : "partIds", "as" : "moduleList"}}] in collection parts
```

Core Feature: Verifying Statistics Endpoint



Verification of Indexes:

